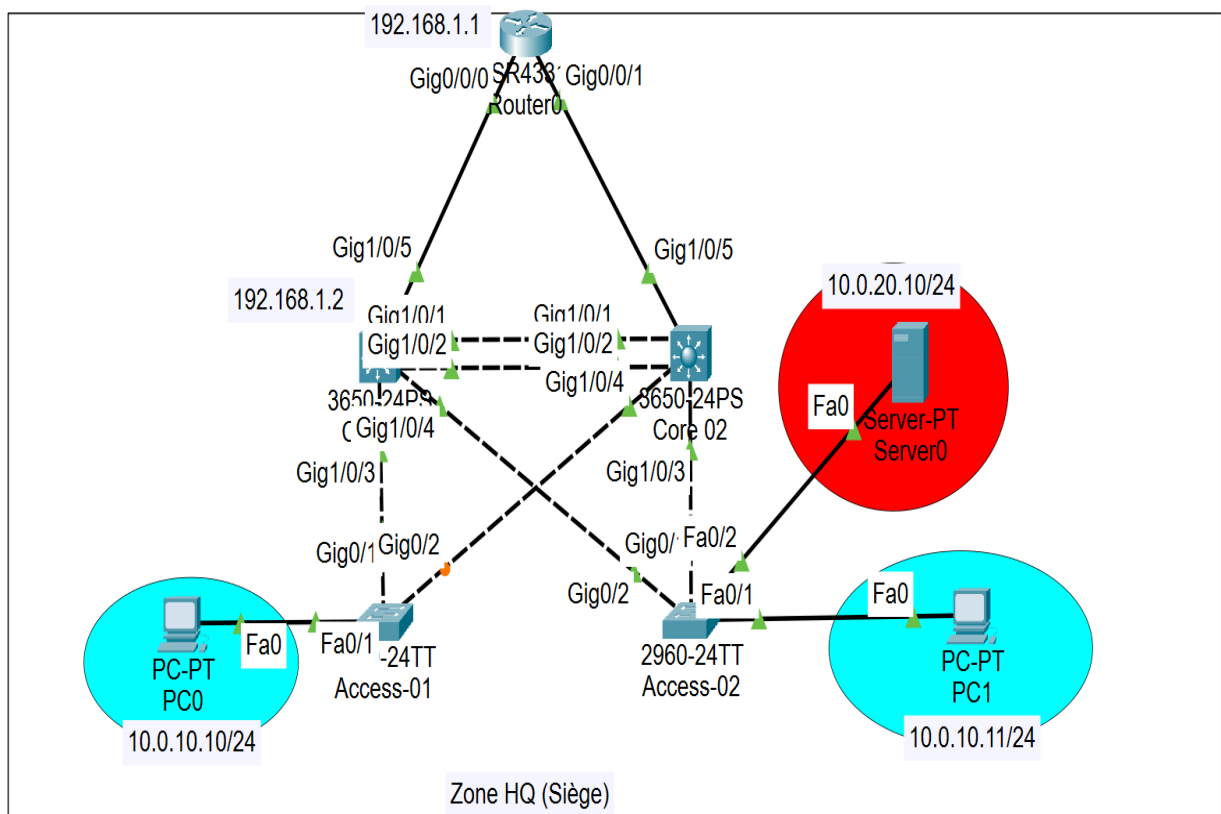


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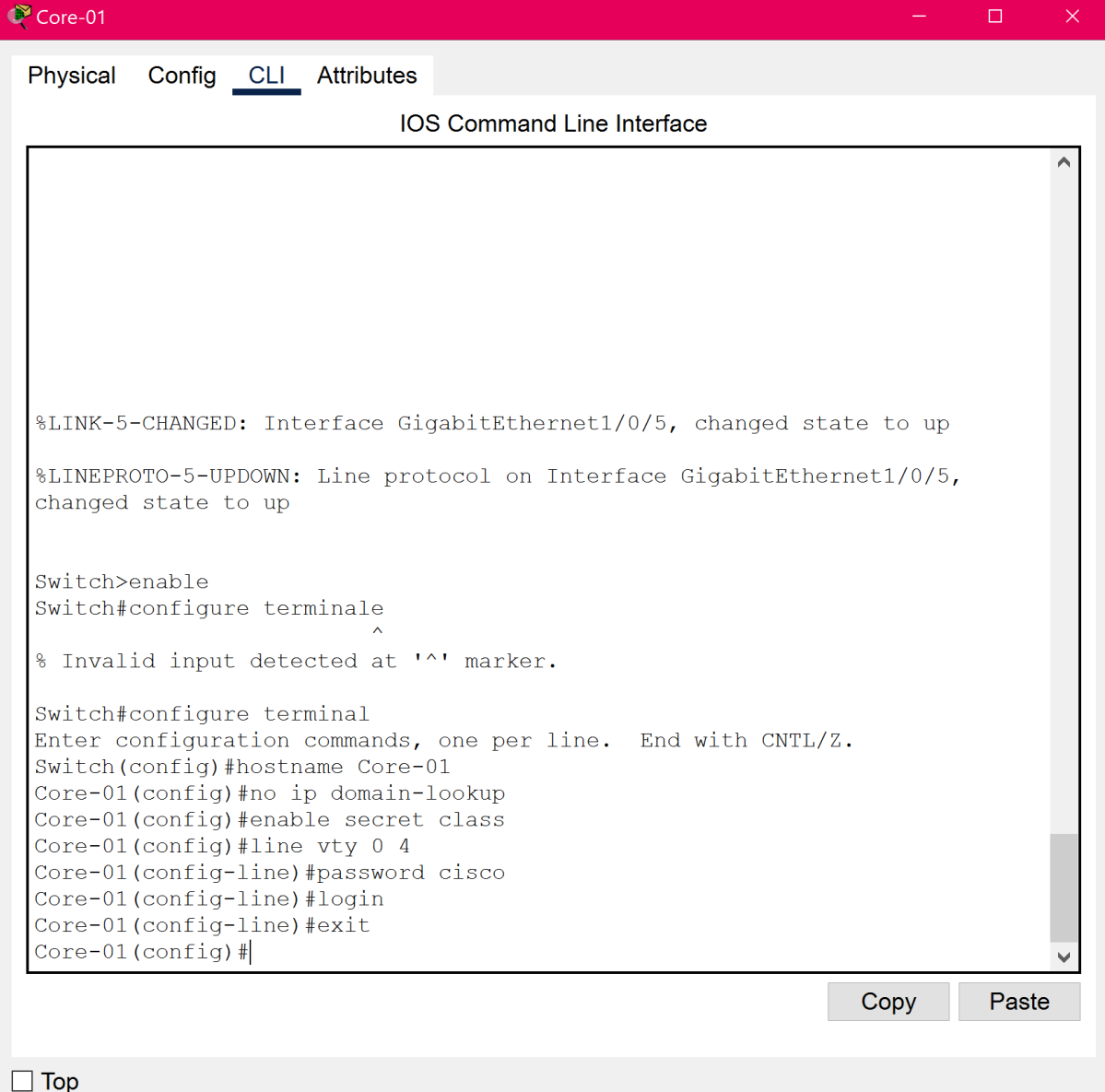
Introduction

Présentation de la topologie réalisée sur Cisco Packet Tracer. L'infrastructure repose sur deux switches de cœur de réseau (Core-01 et Core-02) et deux switches d'accès (Access-01 et Access-02)



Étape 1 : Initialisation et Sécurité

- **Action** : Attribution des noms (hostname) et sécurisation des lignes.
- **Commandes clés** : `enable secret`, `line vty 0 4`, `login local`.



The screenshot shows a terminal window titled 'Core-01' with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the 'IOS Command Line Interface'. The terminal output shows the following sequence of commands and responses:

```
%LINK-5-CHANGED: Interface GigabitEthernet1/0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/5,
changed state to up

Switch>enable
Switch#configure terminale
      ^
% Invalid input detected at '^' marker.

Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Core-01
Core-01(config)#no ip domain-lookup
Core-01(config)#enable secret class
Core-01(config)#line vty 0 4
Core-01(config-line)#password cisco
Core-01(config-line)#login
Core-01(config-line)#exit
Core-01(config)#|
```

At the bottom of the terminal window, there are 'Copy' and 'Paste' buttons, and a 'Top' link.

Étape 2 : Agrégation de liens (EtherChannel LACP)

- **Objectif** : Fusionner les liens entre Core-01 et Core-02 pour éviter les boucles tout en doublant le débit.

Core-02

Physical Config CLI Attributes

IOS Command Line Interface

```
Core-02>en
Password:
Core-02#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Core-02(config)#in
Core-02(config)#interface r
Core-02(config)#interface range g
Core-02(config)#interface range gigabitEthernet1/01/1-2
Core-02(config-if-range)#
Core-02(config-if-range)#channel-group 1 mode active
Core-02(config-if-range)#
Creating a port-channel interface Port-channel 1

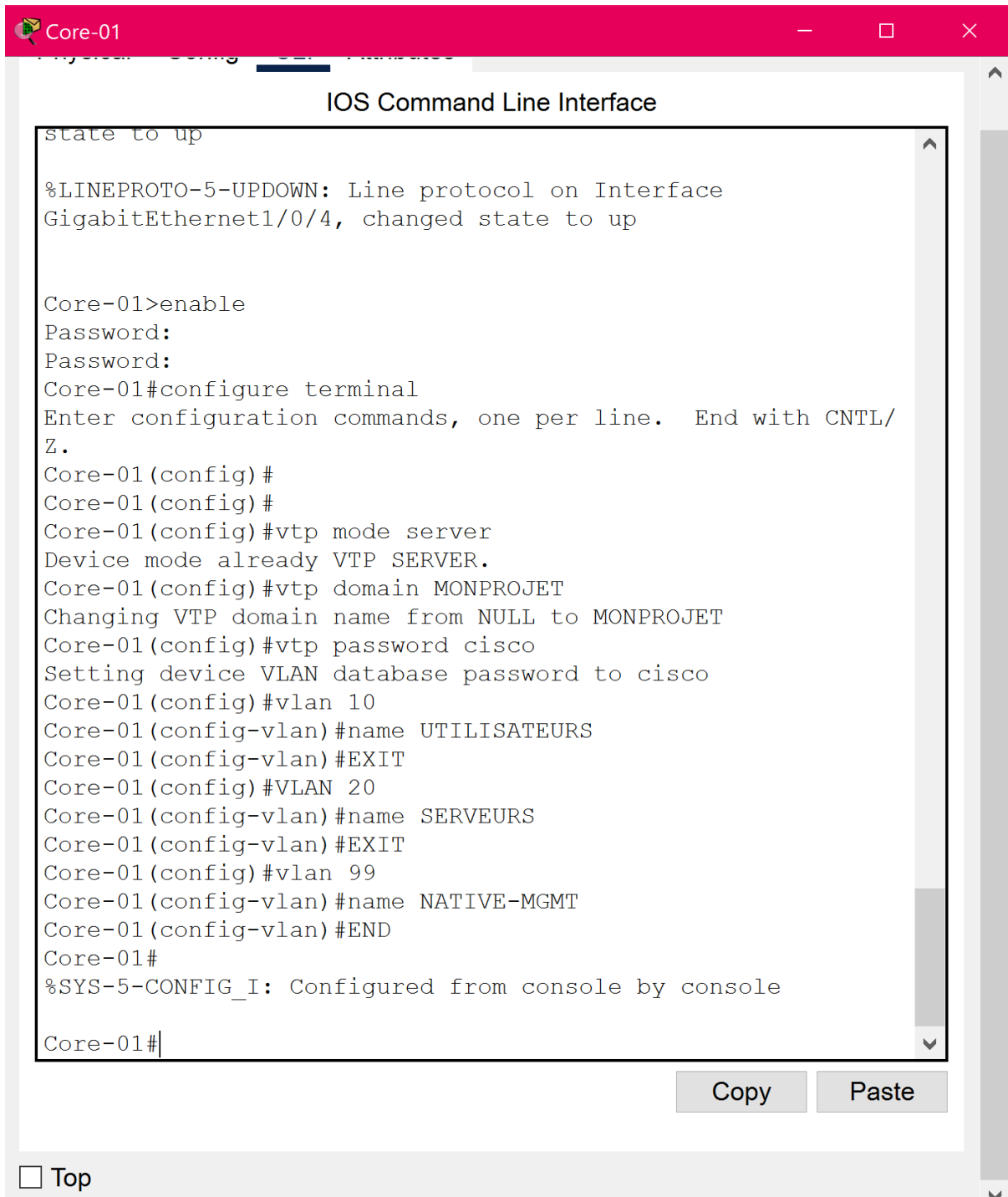
Core-02(config-if-range)#exit
Core-02(config)#interface port-channel 1
Core-02(config-if)#switchport mode trunk
Core-02(config-if)#end
Core-02#
%SYS-5-CONFIG_I: Configured from console by console

Core-02#|
```

Copy Paste

Étape 3 : Déploiement des VLANs et Trunks (VTP)

- **Configuration** : Core-01 est le serveur VTP. Access-01/02 sont clients.
- **Vérification** : Validation de la propagation des VLANs vers l'accès.



The screenshot shows the IOS Command Line Interface for Core-01. The interface is titled "Core-01" and "IOS Command Line Interface". The terminal output shows the following commands and responses:

```
state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet1/0/4, changed state to up

Core-01>enable
Password:
Password:
Core-01#configure terminal
Enter configuration commands, one per line.  End with CNTL/
Z.
Core-01(config)#
Core-01(config)#
Core-01(config)#vtp mode server
Device mode already VTP SERVER.
Core-01(config)#vtp domain MONPROJET
Changing VTP domain name from NULL to MONPROJET
Core-01(config)#vtp password cisco
Setting device VLAN database password to cisco
Core-01(config)#vlan 10
Core-01(config-vlan)#name UTILISATEURS
Core-01(config-vlan)#EXIT
Core-01(config)#VLAN 20
Core-01(config-vlan)#name SERVEURS
Core-01(config-vlan)#EXIT
Core-01(config)#vlan 99
Core-01(config-vlan)#name NATIVE-MGMT
Core-01(config-vlan)#END
Core-01#
%SYS-5-CONFIG_I: Configured from console by console

Core-01#
```

At the bottom of the terminal window, there are "Copy" and "Paste" buttons. Below the terminal window, there is a "Top" button with a checkbox.

Core-02

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3,
changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet1/0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4,
changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1,
changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1,
changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2,
changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2,
changed state to up

Core-02>EN
Password:
Core-02#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Core-02(config)#vtp mode client
Setting device to VTP CLIENT mode.
Core-02(config)#vtp domain MONPROJET
Domain name already set to MONPROJET.
Core-02(config)#vtp password cisco
Setting device VLAN database password to cisco
Core-02(config)#
```

Copy

Paste

Access-01

PhysicalConfigCLIAttributes

IOS Command Line Interface

% Invalid input detected at '^' marker.

Access-01(config)#exit

Access-01#

%SYS-5-CONFIG_I: Configured from console by console

Access-01#

Access-01#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1,
10	UTILISATEURS	active	
20	SERVEURS	active	
99	NATIVE-MGMT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Access-01#

Access-01#

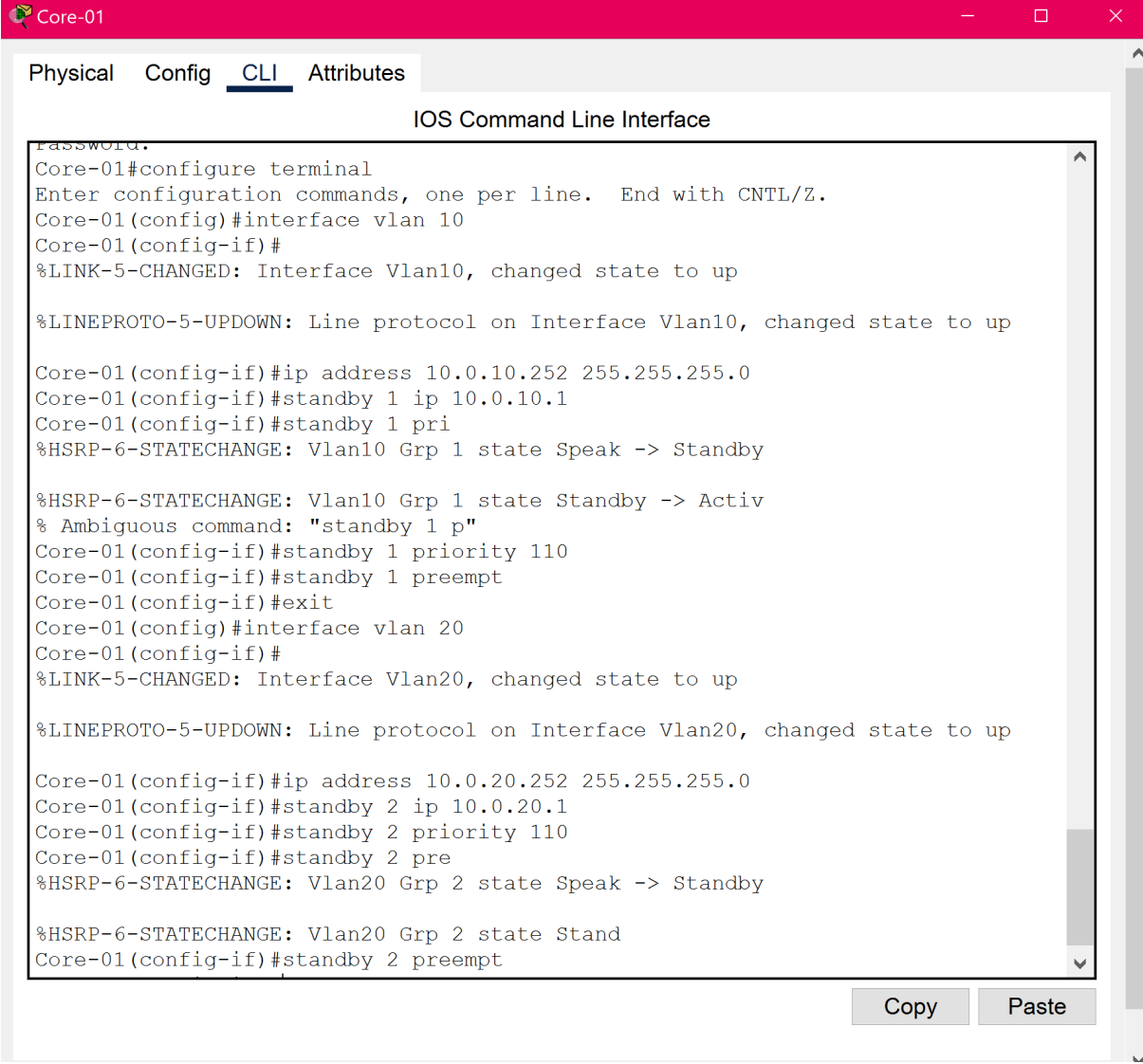
Access-01#

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Paste

Étape 4 : Routage Inter-VLAN et Haute Disponibilité (HSRP)

- **Mécanisme** : Activation de **ip routing**. Création des interfaces VLAN (SVI).
- **HSRP** : Configuration d'une IP virtuelle (VIP) pour que si un Core tombe, l'autre prenne le relais de la passerelle.



The screenshot shows a terminal window titled "Core-01" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following commands and responses:

```
Core-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Core-01(config)#interface vlan 10
Core-01(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
Core-01(config-if)#ip address 10.0.10.252 255.255.255.0
Core-01(config-if)#standby 1 ip 10.0.10.1
Core-01(config-if)#standby 1 pri
%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Standby -> Activ
% Ambiguous command: "standby 1 p"
Core-01(config-if)#standby 1 priority 110
Core-01(config-if)#standby 1 preempt
Core-01(config-if)#exit
Core-01(config)#interface vlan 20
Core-01(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
Core-01(config-if)#ip address 10.0.20.252 255.255.255.0
Core-01(config-if)#standby 2 ip 10.0.20.1
Core-01(config-if)#standby 2 priority 110
Core-01(config-if)#standby 2 pre
%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Stand
Core-01(config-if)#standby 2 preempt
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons.

Core-02

Physical Config CLI Attributes

IOS Command Line Interface

```
Core-02>enable
Password:
Core-02#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Core-02(config)#interface vlan 10
Core-02(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

Core-02(config-if)#ip address 10.0.10.253 255.255.255.0
Core-02(config-if)#standby 1 ip 10.0.10.1
Core-02(config-if)#exit
Core-02(config)#inte
%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Speak -> Stan
Core-02(config)#interface vlan 20
Core-02(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

Core-02(config-if)#ip address 10.0.20.253 255.255.255.0
Core-02(config-if)#standby 2 ip 10.0.20.1
Core-02(config-if)#
%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Speak -> Standby
Core-02(config-if)#
```

Copy Paste

Access-01

```
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed
state to up

Access-01>
Access-01>en
Password:
Access-01#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Access-01(config)#interface f
Access-01(config)#interface fastEthernet 0/3
Access-01(config-if)#switchport mode access
Access-01(config-if)#switchport access vlan 10
Access-01(config-if)#end
Access-01#
%SYS-5-CONFIG_I: Configured from console by console

Access-01#
```

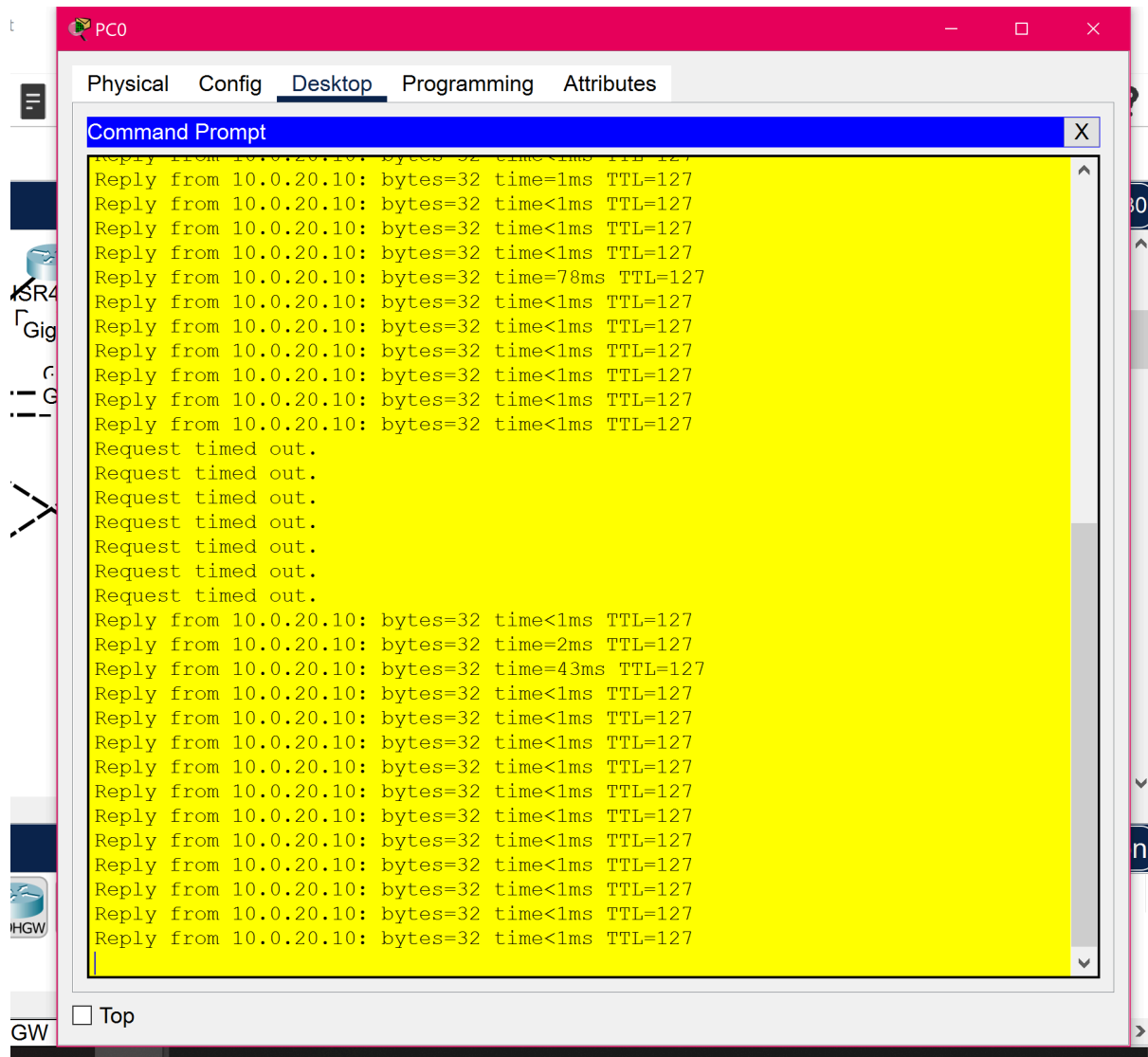
Copy

Paste

☐ Top

Étape 5 : Test de Validation Final

- **Test de Ping** : Vérification de la communication entre PC0 (VLAN 10) et Server0 (VLAN 20).
- même en cas de panne de core-01



```
IOS Command Line Interface

2048K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
250456K bytes of Crash Files at crashinfo : .
1609272K bytes of Flash at flash : .
0K bytes of at webui : .

Base ethernet MAC Address      : 00:02:17:0C:B5:1D
Motherboard assembly number    : 73-15899-06
Motherboard serial number     : FDO20311WHP
Model revision number          : N0
Motherboard revision number    : A0
Model number                   : WS-C3650-24PS
System serial number           : FDO2031Q0TD

Press RETURN to get started!

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed
state to up

%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Standby -> Active

%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Standby -> Active
```

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[□ Top](#)

Core 02

Physical Config CLI Attributes

IOS Command Line Interface

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Standby -> Active

%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Standby -> Active

%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Standby -> Active

%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Standby -> Active

%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Standby -> Active

%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Standby -> Active

Copy Paste

Résultat Finale

